MKOS: A Biomimetic Architecture for Knowledge Base Quality Management

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Abstract

Internal and proprietary knowledge bases—powering enterprise RAG systems, decision support, and organizational memory—accumulate quality issues over time: hallucinations, staleness, bias, and contradictions that degrade downstream reliability. Unlike public knowledge, these issues cannot be caught by an LLM's parametric knowledge alone, as the model has no training data for proprietary content. We present MKOS (Mycelial Knowledge Operating System), a biomimetic architecture that applies biological immune system principles to automated knowledge quality management. MKOS employs a two-phase detection pipeline: (1) an innate classifier that gates healthy content with a single LLM call, and (2) specialized adaptive detectors that perform retrieval-augmented deep analysis for flagged threats. Additional biomimicry subsystems—stigmergic coordination, quorum sensing, and homeostatic self-regulation—enable cross-component signaling and autonomous parameter tuning.

We introduce **KQAB** (Knowledge Quality Assurance Benchmark), a 4-task evaluation suite covering threat detection, contradiction discovery, temporal decay, and evidence grounding, with both synthetic (584 items) and public dataset variants adapted from MNLI and FEVER (923 items total). On KQAB, MKOS achieves Macro-F1=0.975 on threat detection, outperforming an LLM few-shot baseline (F1=0.842) using the same underlying model—with the advantage concentrated on internal-KB-like content where the LLM lacks prior knowledge. An ablation study across 10 configurations confirms that the multi-detector ensemble is essential (innate-only F1=0.139) and that all four specialized detectors contribute measurably. End-to-end evaluation shows that MKOS filtering raises downstream RAG answer correctness from 0.850 to 1.000, matching a clean oracle KB. A complementary knowledge graph approach extracts 1,000 entities and 853 relations to detect structural contradictions across chunks (F1=0.459). We release MKOS as open-source software with reproducible benchmarks.

1 Introduction

Knowledge bases (KBs) are foundational infrastructure for retrieval-augmented generation (RAG), enterprise search, and decision-support systems. As KBs grow, they inevitably accumulate quality issues: factual errors introduced during ingestion (hallucinations), information that becomes outdated as technology evolves (staleness), single-perspective presentations that omit alternatives (bias), and claims that contradict other KB entries or established facts (contradictions).

For *internal* and *proprietary* KBs—company wikis, product documentation, organizational policies—these issues are particularly insidious. An LLM has no parametric knowledge of proprietary content, so naive approaches that rely on the model's training data (e.g., few-shot classification) cannot verify internal claims against ground truth. Quality management requires *retrieval-augmented* verification against the KB itself.

Existing approaches to KB quality management are predominantly manual—expert review, periodic audits, user-reported errors—and scale poorly with KB size. Automated fact-checking systems [Thorne et al., 2018, Guo et al., 2022] focus on verifying individual claims against external corpora but do not address the full spectrum of quality issues within a managed knowledge base. Hallucination detection benchmarks [Li et al., 2023, Lin et al., 2022] evaluate LLM output quality rather than the quality of *stored knowledge*.

We propose MKOS, a system inspired by biological immune systems that treats knowledge quality issues as *pathogens* requiring detection, classification, and remediation. Our key contributions are:

1. **A two-phase immune architecture** that combines fast innate classification (1 LLM call) with deep adaptive analysis by specialized detectors, achieving F1=0.975 on 5-class threat detection.

2. **KQAB**, a benchmark suite for evaluating KB quality systems across four tasks, including two author-constructed tasks (threat detection, temporal decay) and two adaptations of established NLI and fact-verification datasets (MNLI [Williams et al., 2018], FEVER [Thorne et al., 2018]).

3. **Biomimicry subsystems**—stigmergy, quorum sensing, and homeostasis—that enable autonomous cross-component coordination and self-regulation.

4. **End-to-end RAG evaluation** showing that MKOS filtering raises downstream answer correctness by 15 percentage points, and **knowledge graph contradiction detection** as a complementary structural analysis modality.

5. **An ablation study** demonstrating that both the two-phase architecture and the multi-detector ensemble are essential for strong performance.

2 Related Work

Knowledge Base Quality. KB quality management has been studied in the context of knowledge graphs [Paulheim, 2017, Zaveri et al., 2016], where completeness, consistency, and timeliness are primary concerns. These approaches typically rely on structural constraints (ontology validation, type checking) rather than semantic analysis. MKOS targets *unstructured* knowledge bases where content is stored as natural language text.

Automated Fact Checking. Systems like FEVER [Thorne et al., 2018] and fact-verification pipelines [Guo et al., 2022] retrieve evidence and classify claims as supported, refuted, or unverifiable. MKOS extends this paradigm by classifying quality issues into four threat types and operating on *internal* KB content rather than external claims.

Hallucination Detection. Recent work on LLM hallucination detection [Li et al., 2023, Manakul et al., 2023] focuses on identifying unsupported claims in generated text. MKOS applies similar principles to stored knowledge, using retrieval-augmented verification against the KB itself.

Biomimicry in Computing. Artificial immune systems (AIS) have been applied to network intrusion detection [Dasgupta et al., 2011], anomaly detection, and optimization. Stigmergy-based coordination has been explored in swarm robotics and distributed systems [Theraulaz and Bonabeau, 1999]. MKOS adapts these biological metaphors specifically to knowledge quality management.

3 Architecture

MKOS consists of three core subsystems: the Knowledge Immune System (§3.1), biomimicry coordination (§3.2), and the knowledge lifecycle pipeline (§3.3).

3.1 Knowledge Immune System (KIS)

KIS employs a two-phase architecture inspired by biological immunity (Figure **??**):

Phase 1: Innate Response. A single LLM call classifies content into one of five categories: `healthy`, `hallucination`, `staleness`, `bias`, or `contradiction`. The innate classifier uses a structured few-shot prompt with decision rules and examples. If the classifier returns `healthy`, no further analysis occurs—this *gating* mechanism reduces LLM calls by approximately 50%.

Phase 0: Immune Memory. Before adaptive analysis, the system checks for known pathogen patterns in immune memory. Patterns are stored with multi-keyword signatures (requiring 2+ keyword matches, with stop-word filtering to prevent false matches). Matched patterns bypass adaptive analysis, enabling sub-millisecond re-detection.

Phase 2: Adaptive Response. For content flagged as a threat, a specialized detector performs deep analysis:

- **HallucinationDetector**: Retrieves related KB chunks via hybrid search (FTS5 + vector, RRF fusion) and prompts the LLM to compare claims against retrieved evidence.
- **StalenessDetector**: Identifies outdated version references, deprecated technologies, and date-specific claims.
- **BiasDetector**: Detects single-perspective presentation, absolutist language, and missing counterarguments.
- **ContradictionDetector**: Two strategies—FTS-context comparison against related chunks, plus established-facts fallback for isolated contradictions.

Innate-Label Priority. For the ambiguous boundary between bias and contradiction (both involve "opposing" language), the system runs both detectors but *prefers the detector matching the innate label.* If the innate-labeled detector finds threats, the other detector's results are discarded. This optimization raised contradiction F1 from 0.064 to 0.933 by eliminating cross-classification.

3.2 Biomimicry Coordination

Three biological coordination mechanisms enable autonomous system behavior. Table 1 maps each biological metaphor to its concrete software implementation.

Stigmergy. Subsystems communicate indirectly through pheromone signals stored in a shared signal table. Signals carry a type (e.g., `THREAT_DETECTED`), domain, intensity, and source. Repeated signals in the same domain *reinforce*, while all signals *decay* exponentially (72-hour half-life). The fermentation subsystem reads domain threat levels to apply confidence penalties to incoming content in high-threat domains.

Table 1: Biomimicry concepts and their software implementations

Biological Term	Software Implementation	Mechanism
Innate immunity	Few-shot LLM classifier	Fast gate (1 call)
Adaptive immunity	Specialized detector ensemble	Deep analysis
Immune memory	Pattern store + keyword match	Sub-ms re-detection
Stigmergy	Event-driven signal table	Async DB flags w/ decay
Quorum sensing	Threshold-based aggregation	Grouped metric triggers
Homeostasis	PID-like control loop	Setpoint-driven tuning
Composting	Entropy-based archival	$E(t) = 1 - e^{-\lambda \Delta t}$
Fermentation	Cross-ref staging pipeline	Confidence scoring

Quorum Sensing. When the number of unresolved threats in a domain exceeds a configurable threshold (e.g., 3 for hallucination), collective actions are triggered: domain quarantine, cascade composting, batch rescan, or raised scrutiny. This prevents localized quality issues from spreading.

Homeostasis. The system continuously measures vital signs—threat rate, false-positive rate, average entropy, composting throughput, fermentation rejection rate—and compares them against setpoints. When metrics drift outside healthy ranges, the regulator recommends parameter adjustments (e.g., increasing confidence thresholds in high-threat domains).

3.3 Knowledge Lifecycle

Two additional subsystems manage knowledge lifecycle:

Composting. An entropy-based decay model, $E(t) = 1 - \exp(-\lambda \cdot \Delta t)$, scores knowledge freshness. High-entropy chunks are decomposed into reusable *nutrients*—principles, patterns, error patterns, meta-insights—stored with FTS5 indexing for future enrichment.

Fermentation. New content enters a staging chamber, is cross-referenced against existing knowledge via hybrid search, and receives a confidence score based on supporting references and detected contradictions. Content is promoted or rejected based on configurable thresholds.

4 KQAB: Knowledge Quality Assurance Benchmark

Existing benchmarks evaluate either LLM output quality (TruthfulQA, HaluEval) or individual claim verification (FEVER). None evaluate the quality of *stored knowledge* in a managed KB. We introduce KQAB, a 4-task benchmark with three dataset variants.

4.1 Tasks

- **T1: Threat Detection** (5-class). Classify KB chunks as `healthy`, `hallucination`, `staleness`, `bias`, or `contradiction`. Primary metric: Macro-F1.
- **T2: Contradiction Discovery** (3-class). Given a chunk pair, classify as `contradicts`, `consistent`, or `unrelated`. Tests inter-chunk consistency analysis.

- **T3: Temporal Decay Detection** (3-class). Classify KB content as `current`, `outdated`, or `deprecated`. Tests temporal awareness.
- **T4: Evidence Grounding** (3-class). Given a claim and KB context, classify as `supported`, `unsupported`, or `partial`. Tests retrieval-based verification.

4.2 Dataset Variants

KQAB-Synth (584 items). Author-constructed items covering software engineering, data science, and general technical knowledge, designed to simulate quality issues in internal knowledge bases where the LLM has no training data. T1: 321 items with balanced threat-type representation. T3 expanded to 200 items covering both SE and general knowledge domains. All synthetic items and generation code are released for community review and extension.

KQAB-Public (923 items). Adaptations of established NLP datasets for KB-specific evaluation:

- T2: 201 items from Multi-Genre NLI [Williams et al., 2018]. Label mapping: entailment→`consistent`, contradiction→`contradicts`, neutral→`unrelated`. Balanced at 67 items per class.
- T4: 201 items from FEVER [Thorne et al., 2018]. Label mapping: SUPPORTS→`supported`, REFUTES→`unsupported`, NOT_ENOUGH_INFO→`partial`. Balanced at 67 items per class.

T1 and T3 use synthetic items in both variants, as no public dataset exists for KB-specific threat detection or temporal decay classification.

5 Experiments

5.1 Setup

All experiments use Claude Sonnet 4 (Anthropic) as the underlying LLM unless otherwise noted. A SQLite-backed LLM response cache ensures deterministic results across runs ($\sigma = 0.0$). Total benchmark cost: \$5.15 across all experiments.

Systems Compared.

- **MKOS AIS**: Full two-phase immune system with all four adaptive detectors.
- **LLM Few-Shot**: Same LLM with a single few-shot classification prompt. No retrieval, no memory, no specialized detectors.
- **Regex Heuristic**: Keyword-matching baseline.

5.2 T1: Threat Detection Results

Table 2 shows results on the 321-item synthetic dataset.

Per-Class Analysis. Table 3 reveals that MKOS outperforms Few-Shot on *every* threat class, with the largest advantages on bias (+24.8pp) and contradiction (+27.9pp).

Table 2: T1 Threat Detection (n=321, Macro-F1)

System	Precision	Recall	Macro-F1
Regex Heuristic	1.000	0.060	0.113
LLM Few-Shot	0.986	0.953	0.842
MKOS AIS	**0.987**	**0.987**	**0.975**

Table 3: Per-class F1 on T1 (n=321)

Class	AIS F1	Few-Shot F1	Δ
Hallucination	0.980	0.938	+4.2pp
Staleness	0.987	0.919	+6.8pp
Bias	0.983	0.735	+24.8pp
Contradiction	0.933	0.654	+27.9pp
Healthy	0.988	0.966	+2.2pp

5.3 KQAB Multi-Task Results

On KQAB-Public, MKOS consistently matches or outperforms Few-Shot on specialized tasks (T3, T4). The T2 drop from 1.000 to 0.612 on MNLI reflects the greater nuance of real NLI data—both systems struggle equally with the neutral/unrelated class (F1≈0.2).

5.4 Ablation Study

We systematically evaluate component contributions by enabling/disabling detectors (Table 5).

Key findings: (1) The adaptive phase is critical—innate-only achieves F1=0.139, confirming it serves as a gate, not a classifier. (2) All four detectors contribute approximately equally (Δ=0.218–0.253 for leave-one-out). (3) No single detector is sufficient (F1=0.288–0.352 for single-detector configs).

5.5 Multi-Model Evaluation

To test model sensitivity, we evaluated with GPT-4o (Table 6).

The architecture is model-sensitive: Claude AIS (0.975) significantly outperforms GPT-4o AIS (0.869), while raw few-shot performance is comparable across models. This indicates that detector prompts require model-specific tuning.

5.6 Biomimicry Validation

Functional testing of all three biomimicry subsystems achieves 100% pass rate (8/8 tests): stigmergy signal propagation and decay, quorum sensing activation above threshold, and homeostatic parameter diagnosis (see supplementary material).

5.7 Real-World Dataset

A 225-item dataset constructed from Wikipedia article summaries (43 technical topics) with programmatic perturbations shows that Few-Shot (F1=0.976) outperforms AIS (F1=0.963) on *public knowledge*. This is expected: programmatic perturbations to well-known facts are detectable by the

Table 4: KQAB results across all tasks (Macro-F1)

Variant	Task	MKOS	Few-Shot	n
Synth	T1: Threat Detection	**0.975**	0.842	321
	T2: Contradiction	**1.000**	**1.000**	35
	T3: Temporal Decay	**0.872**	0.860	200
	T4: Evidence Grounding	**0.898**	0.867	28
Public	T2: Contradiction (MNLI)	0.612	0.613	201
	T3: Temporal Decay	**0.905**	0.894	200
	T4: Grounding (FEVER)	**0.593**	0.586	201

Table 5: Ablation study on T1 (n=321, 10 configurations)

Configuration	Macro-F1	Δ vs Full
Full AIS (4 detectors)	**0.975**	—
Innate only (no adaptive)	0.139	−0.836
Without hallucination	0.752	−0.223
Without staleness	0.757	−0.218
Without contradiction	0.722	−0.253
Without bias	0.745	−0.230
Only hallucination	0.352	−0.623
Only staleness	0.349	−0.626
Only contradiction	0.314	−0.661
Only bias	0.288	−0.687

LLM's parametric knowledge alone. MKOS's retrieval-based architecture adds the most value on *internal* knowledge bases where the LLM has no prior training data.

5.8 Zero-Shot Enterprise Domain

To test MKOS on truly proprietary content, we constructed a 179-item knowledge base for a *fictional* company ("Nexara Technologies") with five invented products. The LLM has zero training data for this content. Results reveal a critical limitation:

Few-Shot outperforms all MKOS variants on fictional content. Systematic ablation of two proposed fixes reveals a *cascading* failure rather than two independent problems:

Failure Mode 1: Innate Gate Miscalibration. The innate classifier, trained on technical/SE examples, defaults to `healthy` for unfamiliar product names (healthy recall=1.0, threat recall≈0.03). *Automatic calibration*—sampling 50 chunks from the target KB and injecting them as domain-specific few-shot examples—improves F1 from 0.291 to 0.365 (+7.4pp), with bias and staleness detection improving substantially. However, hallucination and contradiction detection remain near zero.

Failure Mode 2: Retrieval Collapse. On KQAB-Synth, 100% of threat items (150/150) find related chunks via hybrid search. On Nexara, the contradiction detector's retrieval finds *zero* of

Table 6: Multi-model evaluation on T1 (n=321)

System	Claude Sonnet 4	GPT-4o
AIS	**0.975**	0.869
Few-Shot	0.842	**0.977**

Table 7: Zero-Shot Enterprise Domain (n=179). Auto-Cal. = automatic calibration with 50 sampled chunks injected as few-shot examples. BM25 = lexical retrieval preferred over vector search for detector context.

System	Macro-F1	Halluc.	Stale.	Bias	Contrad.
MKOS (uncalibrated)	0.291	0.057	0.065	0.764	0.000
MKOS + BM25	0.291	0.057	0.065	0.764	0.000
MKOS + auto-cal.	0.365	0.056	0.333	0.852	0.000
MKOS + auto-cal. + BM25	0.365	0.056	0.333	0.852	0.000
Few-Shot	**0.604**	**0.667**	**0.511**	**0.984**	**0.182**

30 known counterpart chunks (MRR=0.000, Recall@5=0.000). The embedding model (BAAI/bge-small-en-v1.5) has no representations for fictional product names. We hypothesized that BM25 (lexical matching) would perform better, since "QuantumBridge" as a keyword appears in both contradicting and healthy chunks.

Critical Finding: Cascading, Not Independent. Adding BM25-preferred retrieval alone produces *zero improvement* (+0.0pp); combining it with calibration also yields no additional gain beyond calibration alone. The reason: the innate gate blocks threats *before* retrieval occurs. Even with perfect retrieval, chunks classified as "healthy" by the gate never reach the adaptive detectors. The two failure modes are *cascading*, not independent—the gate failure dominates, and retrieval fixes can only take effect after the gate is opened. This has a direct architectural implication: for zero-shot domains, the innate gate should be bypassed or relaxed (e.g., by lowering the healthy-confidence threshold), allowing all chunks to reach adaptive detectors at the cost of more LLM calls.

5.9 End-to-End RAG Impact

To measure MKOS's downstream utility, we evaluate answer quality in a retrieval-augmented generation (RAG) pipeline under three conditions: (1) *Degraded*—the KB contains both healthy and threat chunks without filtering; (2) *Filtered*—MKOS scans and removes detected threats before RAG retrieval; (3) *Clean*—an oracle KB containing only healthy chunks. We construct a 30-item KB across 10 technical domains (Python, Docker, SQL, Kubernetes, REST, Git, ML, Security, Caching, Testing), each with 2 healthy chunks and 1 threat. An LLM-as-judge evaluates correctness (factual accuracy) and faithfulness (grounding in retrieved context) for 10 domain-specific questions.

MKOS filtering eliminates all 10/10 threat chunks (detection rate=100%), raising correctness from 0.850 to 1.000 (+15pp) and matching the clean oracle condition. This demonstrates that upstream quality filtering has a measurable, practical impact on downstream answer reliability.

Table 8: End-to-end RAG quality under three KB conditions

Metric	Degraded	MKOS-Filtered	Clean Oracle
Correctness	0.850	**1.000**	1.000
Faithfulness	0.880	**1.000**	1.000
Threat usage rate	100%	0%	0%

5.10 Knowledge Graph Contradiction Detection

While the per-chunk immune system classifies individual items in isolation, some contradictions are only detectable through *structural* analysis across chunks. We build an entity-relation knowledge graph over the full T1 dataset (321 chunks) by extracting subject-predicate-object triples via LLM, normalizing entity names to canonical forms, and connecting chunks that share entities.

The graph contains 1,000 entities, 853 relations, and 15 contradiction clusters (connected components of mutually contradicting chunks). Direct contradiction detection—comparing relations about the same subject from different chunks, verified by LLM—finds 26 contradictions. Against T1 ground truth:

Table 9: Knowledge graph contradiction detection on T1 (n=321)

Metric	Value
Entities extracted	1,000
Relations extracted	853
Direct contradictions found	26
Contradiction clusters	15
Precision (vs. ground truth)	0.452
Recall (vs. ground truth)	0.467
F1	0.459
Threat overlap rate	51.6%

The knowledge graph provides a complementary detection modality: while per-chunk classification achieves higher F1 (0.975), the graph identifies *structural* contradiction patterns (e.g., "unit tests must test private methods" in one chunk vs. "unit tests should test the public interface" in another) that emerge from cross-chunk entity alignment rather than single-item analysis. The 51.6% threat overlap rate indicates that graph-flagged chunks are more likely to be genuine threats than random chunks.

5.11 Uncertainty Quantification

Inspired by SelfCheckGPT [Manakul et al., 2023], we implement self-consistency sampling: the innate classifier is run $N = 5$ times at temperature > 0 for each chunk, and label agreement rates are used to calibrate confidence. On a 25-item stratified sample from T1, all items achieve 100% agreement across samples, indicating that the synthetic dataset is well within the classifier's confidence boundary. The calibrated confidence ($\text{conf}_{\text{cal}} = \text{conf}_{\text{orig}} \times \text{agreement}$) enables prioritization for human review: items with $< 80\%$ agreement are flagged as uncertain, supporting a human-in-the-loop workflow where review effort is concentrated on borderline cases.

6 Discussion

Why Architecture Matters. The 13.3pp F1 advantage of MKOS over Few-Shot (0.975 vs. 0.842) on T1 demonstrates that structured, multi-phase detection outperforms monolithic LLM classification. The advantage is most pronounced for bias (+24.8pp) and contradiction (+27.9pp), where specialized prompts and evidence retrieval add the most value.

Innate-Label Priority. A key design insight is that the innate classifier, while insufficient alone (F1=0.139), provides a reliable *category signal* for routing. When the bias and contradiction detectors both fire, preferring the innate-labeled result eliminates cross-classification—improving contradiction F1 from 0.064 to 0.933.

Domain Sensitivity Spectrum. MKOS performance varies dramatically across domain types, revealing a three-regime pattern: (1) On the *calibrated* KQAB-Synth domain, MKOS dominates (F1=0.975 vs. 0.842). (2) On *public knowledge* (Wikipedia), Few-Shot marginally wins (0.976 vs. 0.963) because the LLM's parametric knowledge suffices. (3) On a *zero-shot fictional* domain (Nexara Technologies), Few-Shot dominates (0.604 vs. 0.282) because MKOS's innate classifier is uncalibrated. This reveals that MKOS's advantage is *domain-conditional*: the system excels when calibrated on the target domain, but requires a calibration phase for new domains. The few-shot baseline, while weaker on calibrated domains, degrades more gracefully to unseen content.

Biomimicry: Metaphor vs. Mechanism. We acknowledge that the biological terminology (stigmergy, quorum sensing, homeostasis) maps onto well-understood software patterns. Stigmergy implements an event-driven message bus with exponential decay; quorum sensing is threshold-based aggregation over grouped metrics; homeostasis is a PID-like control loop with configurable setpoints. We retain the biological framing because it provides a *coherent design vocabulary* that guided architectural decisions—e.g., the two-phase innate/adaptive split directly mirrors immunological design and motivated the gating mechanism that proved essential (innate-only F1=0.139 vs. full F1=0.975). The metaphor is productive, not merely decorative, but readers should understand the concrete mechanisms behind each term.

Methodological Independence. A potential circularity concern arises from the fact that Claude (Anthropic) serves as the underlying LLM in MKOS *and* assisted during architecture design, code implementation, and manuscript drafting. If the synthetic T1 dataset were implicitly tuned to Claude's latent representations, the reported F1=0.975 could reflect model–data co-adaptation rather than genuine architectural merit. We address this concern with five lines of evidence.

First, the *cross-model Few-Shot result* is consistent with non-contaminated data: GPT-4o Few-Shot achieves F1=0.977 on the same synthetic T1 dataset where Claude Few-Shot scores only 0.842. If the data were co-adapted to Claude's representations, we would expect Claude to outperform GPT-4o—not underperform by 13.5pp. We note, however, that this observation is not conclusive: GPT-4o may simply be a stronger few-shot classifier, while Claude benefits from prompt-tuned multi-detector routing. These effects (model capability vs. data bias) are not fully disentangled in our evaluation.

Second, on independently constructed public datasets (MNLI-derived T2, FEVER-derived T4), MKOS performs comparably to Few-Shot (T2: 0.612 vs. 0.613; T4: 0.593 vs. 0.586), showing no systematic advantage from using the same model. Third, on the zero-shot fictional domain (§5.8), MKOS *underperforms* Few-Shot by 32.2pp—the opposite of what contamination would predict.

Fourth, the MKOS performance drop on GPT-4o (F1: 0.975→0.869) while Few-Shot remains stable indicates *prompt-calibration* sensitivity: the detector prompts are tuned to Claude's instruction-following style, not to Claude-specific data patterns. This is a limitation of prompt engineering, not evidence of data contamination.

Fifth, all 321 T1 items were hand-authored as readable natural-language paragraphs, not generated by any LLM. The full dataset, all source code, detector prompts, and cached LLM responses are released under an open-source license, enabling independent reproduction and forensic audit of every classification decision. We nevertheless recommend that future work include crowd-sourced inter-annotator agreement (Cohen's $\kappa \geq 0.80$) on a representative subset to provide independent validation of label quality.

6.1 Limitations

L1: Dataset Size and Author Bias. The primary T1 result (F1=0.975) rests on 321 author-constructed synthetic items without independent crowd-sourced annotation or inter-annotator agreement measurement (e.g., Cohen's κ). This creates a risk of overfitting to the author's implicit model of what constitutes each threat type. We mitigate this concern in three ways: (1) KQAB includes public-dataset variants adapted from established NLI (MultiNLI, n=201) and fact-verification (FEVER, n=201) benchmarks that are entirely independent of the author; (2) the ablation study's consistent pattern across 10 configurations suggests the architecture generalizes beyond individual items; and (3) we release all datasets and code to enable community validation and extension. Future work should pursue three directions: LLM-assisted data synthesis to scale to 2,000–5,000 items with programmatic perturbation injection; crowd-sourced inter-rater reliability measurement on a subset of 200+ items; and expanded evaluation on fully fictional "zero-shot" enterprise domains, building upon our initial Nexara Technologies findings (§5.8) which revealed the innate gate's domain-sensitivity limitations.

L2: Scalability and Cost. Each scan requires 1–3 LLM API calls at approximately \$0.005 per item. For an enterprise KB with 100K chunks, a full scan would cost approximately \$500—prohibitive for routine monitoring. Three mitigation strategies are viable: (1) *Knowledge distillation*—using MKOS scan results as training data to fine-tune a small local model (e.g., Llama-3-8B) for Phase 1 innate classification, reducing the per-item cost to near zero for the gating phase; (2) *Delta scanning*—operating event-driven rather than batch, scanning only newly ingested chunks and their semantic neighbors, consistent with the biological metaphor of immune surveillance at entry points; and (3) the response caching layer already implemented, which makes repeat scans cost \$0. We view the current work as validating the *architecture*, with cost optimization as engineering follow-up.

L3: Model Sensitivity and Prompt Fragility. MKOS performance drops significantly on GPT-4o (AIS F1: 0.975→0.869) while the Few-Shot baseline remains stable across models (0.842 vs. 0.977). This indicates that the detector prompts are tuned to Claude Sonnet 4's instruction-following characteristics rather than being model-agnostic. The architecture itself (two-phase detection, evidence retrieval, ensemble) is model-independent, but the specific prompts constitute a form of soft engineering that requires per-model calibration. We frame this as a *calibration phase*: analogous to how biological immune systems undergo thymic education, MKOS requires prompt tuning against a small gold-standard dataset when initialized with a new base LLM. Future work should explore automated prompt optimization frameworks (e.g., DSPy) to reduce this calibration effort.

L4: Domain Coverage. All evaluation focuses on technical and software engineering knowledge. Generalization to domains with different epistemological characteristics—medical knowledge (where "contradiction" has different semantics), legal text (where "bias" is context-dependent), or rapidly evolving fields—requires domain-specific evaluation and likely detector adaptation.

L5: Biomimicry Evaluation Depth. The biomimicry subsystems (stigmergy, quorum sensing, homeostasis) are validated through functional tests (8/8 pass) and a longitudinal simulation. We simulate 8 waves of KB growth (64 items total) with progressively subtler threats clustering in a database domain, comparing full MKOS (with domain-aware alertness fed by stigmergy signals) against stateless detection. Both modes achieve F1=1.000 on every wave—a *ceiling effect* caused by Claude Sonnet 4's strong innate classification, even for subtle threats designed to be ambiguous. Importantly, the biomimicry mechanics are functional: stigmergy signals accumulate from 3 to 13 active signals over 8 waves, quorum sensing fires at wave 4 when DB-domain threats exceed the threshold, and homeostasis adjusts parameters throughout. The ceiling effect prevents us from demonstrating that these coordination mechanisms improve detection accuracy on LLM-readable text. Their value may emerge in deployments with weaker base models, noisier domains, or extended operation timescales beyond the scope of this evaluation.

L6: Baseline Diversity. We compare MKOS against only two baselines: an LLM few-shot classifier and a regex heuristic. Notably absent are chain-of-thought prompting, fine-tuned classifiers (e.g., a BERT model trained on our T1 labels), and existing fact-checking pipelines applied to KB data. The 13.3pp advantage over few-shot may narrow or disappear against stronger baselines. We prioritized the few-shot comparison because it isolates the architectural contribution (same LLM, different pipeline), but acknowledge that a broader baseline sweep is necessary for definitive claims about MKOS's relative merit.

7 Conclusion

We presented MKOS, a biomimetic architecture for knowledge base quality management that draws on immune system principles, stigmergic coordination, quorum sensing, and homeostatic self-regulation. On the author-constructed KQAB-Synth benchmark, MKOS achieves Macro-F1=0.975 on 5-class threat detection, outperforming an LLM few-shot baseline by 13.3 percentage points. An ablation study confirms that both the two-phase architecture and all four specialized detectors contribute measurably. End-to-end RAG evaluation demonstrates practical impact: MKOS filtering raises downstream answer correctness from 0.850 to 1.000 (+15pp), matching a clean oracle KB. A complementary knowledge graph extracts entity-relation triples to detect structural contradictions across chunks (F1=0.459, 15 clusters), providing a second detection modality alongside per-chunk classification. These results are preliminary: they rest on a small, single-annotator dataset (n=321) and require independent validation through crowd-sourced annotation and broader baseline comparisons before definitive claims can be made.

MKOS provides initial evidence that structured, retrieval-augmented architectures can complement monolithic LLM approaches for knowledge quality management, particularly for internal knowledge bases where the LLM lacks training data coverage. However, the system's dependence on domain calibration (§5.8) and model-specific prompt tuning (L3) currently limits its out-of-the-box applicability.

MKOS and KQAB are released as open-source software to support reproducibility and community benchmarking.

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